Tech Dessert #7: Electrical and Tower Safety

Companion reference for meeting 7 -- staying safe around AC power, batteries, lightning, and antenna towers.

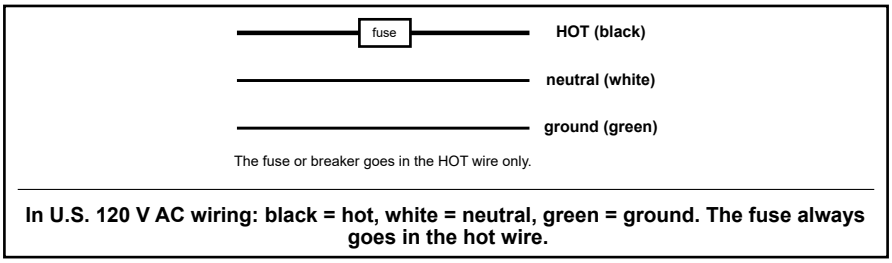
Groups T0A + T0B · 23 pool questions · Self-study handout · Practice exams at eham.net/exams

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This Dessert is the most important one to take seriously – it is about not getting hurt. It is a reference sheet: read the tables, study the diagrams, then drill the question bank.

Electrical Safety

Topic	What to know	Q
Biggest hazard of an unprotected 12 V battery	**Shorting the terminals** can cause burns, fire, or an explosion.	T0A01
How electrical current harms the body	It heats tissue, disrupts cell function, and causes muscle contractions – **all of these**.	T0A02
What black wire insulation means (120 V AC)	**Hot.**	T0A03
Purpose of a fuse	To **remove power** in case of an overload.	T0A04
Why never replace a 5 A fuse with a 20 A fuse	Excessive current could cause a **fire**.	T0A05
Where a fuse or breaker goes in a 120 V AC circuit	In series with the **hot conductor only**.	T0A08
Guarding against electrical shock	Three-wire cords, a common safety ground, and discharging high-voltage capacitors – **all of these**.	T0A06
Hazard right after a power supply is switched off	Charge still stored in the **filter capacitors**.	T0A11
Hazard of rapidly charging/ discharging a battery	**Overheating or out-gassing.**	T0A10
Where a lightning arrester goes in a feed line	On a **grounded panel** where the feed lines enter the building.	T0A07
What to do with all external ground rods	**Bond them together** with heavy wire or strap.	T0A09
Measuring high voltage safely	Use a voltmeter and leads **rated for that voltage**.	T0A12

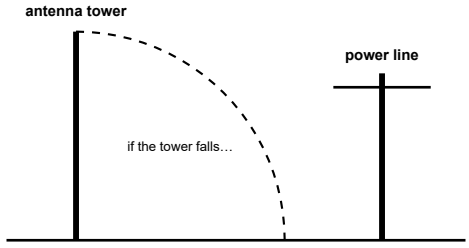


In U.S. 120 V AC wiring: black = hot, white = neutral, green = ground. The fuse always goes in the hot wire.

Antenna and Tower Safety

Topic	What to know	Q
Climbing an antenna tower requires	Training, a tie-off at all times, and an approved harness – **all of these**.	T0B02
Climbing a tower alone	**Never** safe – always have a helper or observer.	T0B03
Main danger when putting up a tower	**Overhead electrical wires** – look for them and stay clear.	T0B04
Minimum safe distance from a power line	Far enough that the antenna, if it falls, cannot come within **10 feet** of the power wires.	T0B06
A crank-up tower	Do not climb it unless it is **retracted** or its safety locks are set.	T0B07
Attaching an antenna to a utility pole	Avoid it – the antenna could **contact high-voltage power lines**.	T0B09
Purpose of a safety wire through a turnbuckle	Keeps the turnbuckle from **loosening due to vibration**.	T0B05
Proper way to ground a tower	Separate **eight-foot ground rods for each leg**, bonded together.	T0B08
Tower ground wires for lightning protection	Keep the connections **short and direct**.	T0B01
Routing lightning grounding conductors	**Avoid sharp bends.**	T0B10

What sets grounding requirements for a tower	**Local electrical codes.**	T0B11



Keep enough distance that a falling tower stays 10 ft clear of the wires.

Never put up an antenna where it could reach a power line if it fell – keep a 10-foot margin.

Remember – this Dessert is about not getting hurt: never short a battery; the **fuse goes in the hot wire**; a power supply holds dangerous **charge in its capacitors** even after it is off. Never climb a tower **alone** or without a harness, and never put up an antenna where it could fall into a **power line**. When in doubt, stop and get help – no contact is worth an injury.

Dessert 7 Question Bank (T0A + T0B)

T0A01. Which of the following is a safety hazard of a 12-volt storage battery that lacks internal protection circuitry?
A. Touching both terminals with your hands can cause electrical shock
B. Shorting the terminals can cause burns, fire, or an explosion
C. RF emissions from a nearby transmitter can cause the electrolyte to emit poison gas
D. All these choices are correct

T0A02. What health hazard is posed by electrical current flowing through the body?
A. It may cause injury by heating body tissue
B. It may disrupt the electrical functions of cells
C. It may cause involuntary muscle contractions
D. All these choices are correct

T0A03. In the United States, what circuit does black wire insulation indicate in a three-wire 120 V AC cable?
A. Neutral
B. Hot
C. Equipment ground
D. Negative

T0A04. What is the purpose of a fuse in an electrical circuit?
A. To prevent power supply ripple from damaging a component
B. To remove power in case of an overload
C. To limit current and prevent shocks
D. All these choices are correct

T0A05. Why should a 5-ampere fuse never be replaced with a 20-ampere fuse?
A. The larger fuse would be likely to blow because it is rated for higher current
B. The power supply ripple would greatly increase
C. Excessive current could cause a fire
D. Voltage drop in the higher current fuse could result in excessively low voltage to the device

T0A06. What is a good way to guard against electrical shock at your station?
A. Use three-wire cords and plugs for all AC powered equipment
B. Connect all AC powered station equipment to a common safety ground
C. Ensure all capacitors used for high-voltage DC are fully discharged before working inside equipment
D. All these choices are correct

T0A07. Where should a lightning arrester be installed in a coaxial feed line?
A. At the output connector of a transceiver
B. At the antenna feed point
C. At the AC power service panel
D. On a grounded panel near where feed lines enter the building

T0A08. Where should a fuse or circuit breaker be installed in a 120V AC power circuit?
A. In series with the hot conductor only
B. In series with the hot and neutral conductors
C. In parallel with the hot conductor only
D. In parallel with the hot and neutral conductors

T0A09. What should be done to all external ground rods or earth connections?
A. Waterproof them with silicone caulk or electrical tape
B. Keep them as far apart as possible
C. Bond them together with heavy wire or conductive strap
D. Tune them for resonance on the lowest frequency of operation

T0A10. What hazard exists when rapidly charging or discharging an unprotected battery?
A. Overheating or out-gassing
B. Excess output ripple
C. Electric shock
D. Overvoltage

T0A11. What hazard exists in a power supply immediately after turning it off?
A. Circulating currents in the dc filter
B. Leakage flux in the power transformer
C. Voltage transients from kickback diodes
D. Charge stored in filter capacitors

T0A12. Which of the following precautions should be taken when measuring high voltages with a voltmeter?
A. Ensure that the voltmeter has very low impedance
B. Ensure that the voltmeter and its leads are rated for use at the voltages being measured
C. Ensure that the circuit is grounded through the voltmeter
D. Ensure that the voltmeter is set to the correct frequency

T0B01. Which of the following is good practice when installing ground wires on a tower for lightning protection?
A. Put a drip loop in the ground connection to prevent water damage to the ground system
B. Make sure all ground wire bends are right angles
C. Ensure that connections are short and direct
D. All these choices are correct

T0B02. What is required when climbing an antenna tower?
A. Have sufficient training on safe tower climbing techniques
B. Use appropriate tie-off to the tower at all times
C. Always wear an approved climbing harness
D. All these choices are correct

T0B03. Under what circumstances is it safe to climb a tower without a helper or observer?
A. When no electrical work is being performed
B. When no mechanical work is being performed
C. When the work being done is not more than 20 feet above the ground
D. Never

T0B04. Which of the following is an important safety precaution to observe when putting up an antenna tower?
A. Wear a ground strap connected to your wrist at all times
B. Insulate the base of the tower to avoid lightning strikes
C. Look for and stay clear of any overhead electrical wires
D. All these choices are correct

T0B05. What is the purpose of a safety wire through a turnbuckle used to tension guy lines?
A. Secure the guy line if the turnbuckle breaks
B. Prevent loosening of the turnbuckle from vibration
C. Provide a ground path for lightning strikes
D. Provide an ability to measure for proper tensioning

T0B06. What is the minimum safe distance from a power line to allow when installing an antenna?
A. Add the height of the antenna to the height of the power line and multiply by a factor of 1.5
B. The height of the power line above ground
C. 1/2 wavelength at the operating frequency
D. Enough so that if the antenna falls, no part of it can come within 10 feet of the power wires

T0B07. Which of the following is an important safety rule to remember when using a crank-up tower?
A. This type of tower must never be painted
B. This type of tower must never be grounded
C. This type of tower must not be climbed unless it is retracted, or mechanical safety locking devices have been installed
D. All these choices are correct

T0B08. Which is a proper grounding method for a tower?
A. A single four-foot ground rod, driven into the ground no more than 12 inches from the base
B. A ferrite-core RF choke connected between the tower and ground
C. A connection between the tower base and a cold-water pipe
D. Separate eight-foot ground rods for each tower leg, bonded to the tower and each other

T0B09. Why should you avoid attaching an antenna to a utility pole?
A. The antenna will not work properly because of induced voltages
B. The antenna may unbalance the power transformer, causing power fluctuations
C. The antenna could contact high-voltage power lines
D. All these choices are correct

T0B10. Which of the following is true when installing grounding conductors used for lightning protection?
A. Use only non-insulated wire
B. Wires must be carefully routed with precise right-angle bends
C. Sharp bends must be avoided
D. Common grounds must be avoided

T0B11. Which of the following establishes grounding requirements for an amateur radio tower or antenna?
A. FCC Part 97 rules
B. Local electrical codes
C. FAA tower lighting regulations
D. UL recommended practices

Answer Key

T0A01: B	T0A02: D	T0A03: B	T0A04: B	T0A05: C
T0A06: D	T0A07: D	T0A08: A	T0A09: C	T0A10: A
T0A11: D	T0A12: B	T0B01: C	T0B02: D	T0B03: D
T0B04: C	T0B05: B	T0B06: D	T0B07: C	T0B08: D
T0B09: C	T0B10: C	T0B11: B		